

Web User Experience & Database Design

User Interface Principles & Design



By Joop

Inthraporn Aranyanak, PhD

Lecture 3- Key Topics

- The Psychology of color
- Typography in UI design
- From Flows to Prototypes

- What is GUI?
- Gestalt Theory
- Metaphor in UI Design



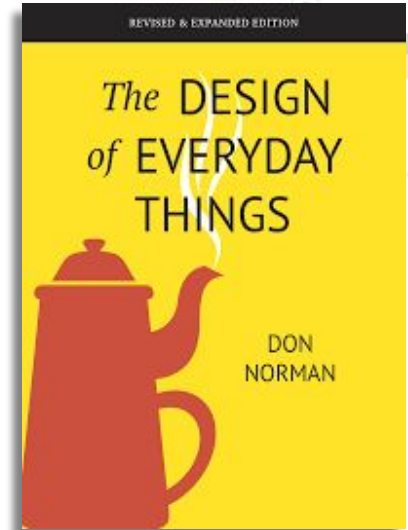


[CrossList-Web User](#)
[Exp Data Design-](#)
[WADT-H1002-Sept-](#)
[Dec Academic Term](#)
[2026/27-4](#)

“Make sure that you can access this module on Brightspace by the end of Week 5, since I will hand out CA1 in Week 6 and you will have to submit your work on Brightspace in Week 11.”

Design principles by Donald Norman (2013)

1. What is **Visibility** , and what are some examples?
2. What is **Feedback** , and what are some examples?
3. What is **Affordance** , and what are some examples?
4. What is **Signifier** , and what are some examples?
5. What is **Mapping** , and what are some examples?
6. What is **Consistency** , and what are some examples?
7. What is **Constraint** , and what are some examples?



If you have any challenges (such as colour blindness, ADHD, or dyslexia) that might affect your design mockups, please explain them in your report when you submit your CAI.

Note: you can design based on your own experience — just make sure you have rational reasons and principles to support your design choices.

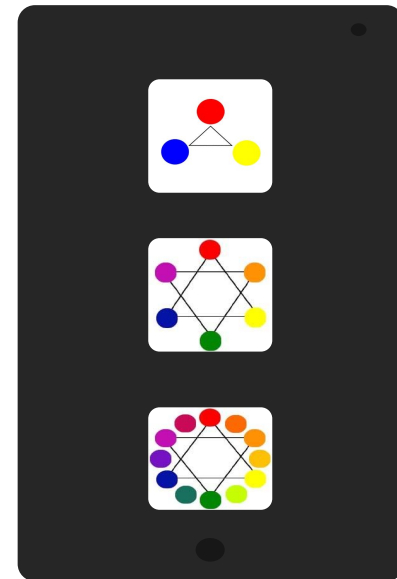


The Psychology of Color

Color is a key element in design, from art and advertising to websites and apps. The right use of color makes users feel comfortable and enhances their experience. The role of color has two main aspects:

- (1) **Psychological and emotional** – colors affect mood and carry cultural meanings (e.g., red = excitement, blue = trust & reliability, pink = love, etc).
- (2) **Functional** – Colors are used to highlight key points, distinguish information, and guide users through the interface.

Understanding both helps designers create work that is attractive and effective.

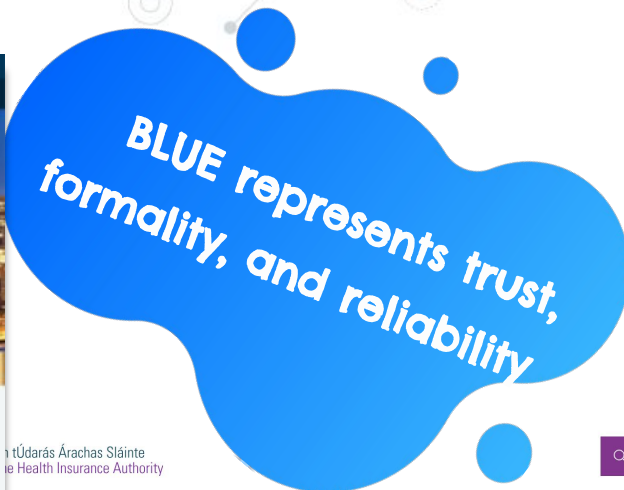


The Psychology of Color

RED	PINK	PURPLE	NAVY	GREEN	BLUE	ORANGE
MEANS: passionate active EXCITING bold energy youthful physical PIONEERING leader winner confidence ambition POWER	MEANS: love calm respect WARMTH longterm feminine intuitive care assertive sensitive NURTURE possibilities UNCONDITIONAL	MEANS: DEEP creativity unconventional original stimulation individual WEALTHmodesty compassion DISTINGUISHED respectable fantasy	MEANS: trust order LOYALTY sincere authority communication confidence PEACE integrity control responsible success CALM masculine	MEANS: BALANCE growth restore sanctuary EQUILIBRIUM positivityNATURE generous clarity prosperity good judgement safetystable	MEANS: spirit perspective CONTENT control rescue determination self-sufficient modern goals awarePURPOSE OPEN ambition	MEANS: INSTINCT WARMTH gut reaction optimistic spontaneity extrovert socialNEW IDEAS FREEDOM impulse motivation
BRANDS: Kellogg's Virgin LEGO Coca-Cola Nintendo Red Bull P	BRANDS: BBC three Barbie COSMOPOLITAN VICTORIA'S SECRET	BRANDS: Cadbury YAHOO! Hallmark Mika Zoopl.co.uk Your advantage is property	BRANDS: f Reebok British Gas t GAP	BRANDS: bp Holiday Inn tic tac LACOSTE Starbucks	BRANDS: e (intel) Blu-ray Disc skype Twitter W	BRANDS: fanta orange Penguin MasterCard bitly B

The Psychology of Color

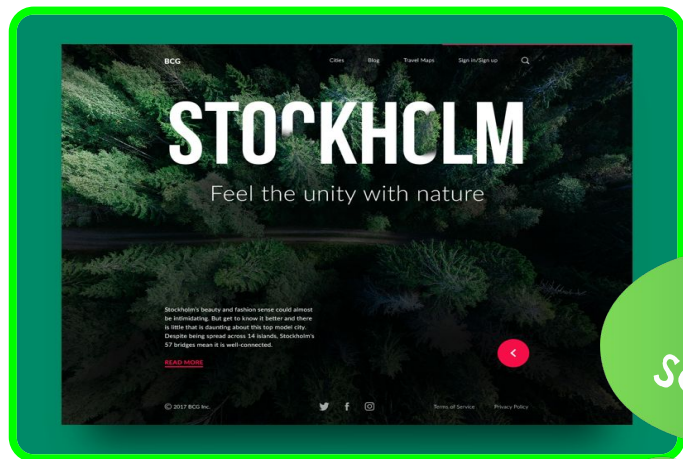




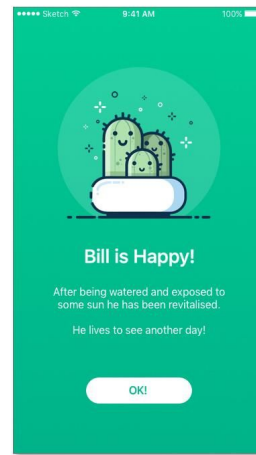
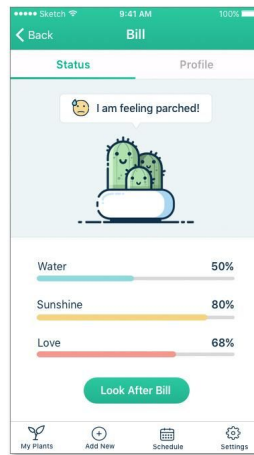
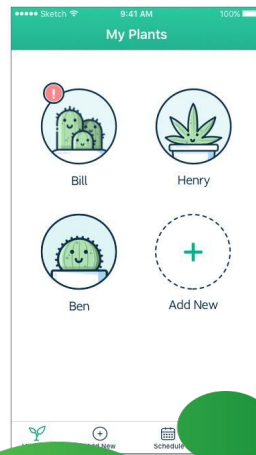
tÚdarás Árachas Sláinte
Health Insurance Authority

Helpline: (01) 406 0080

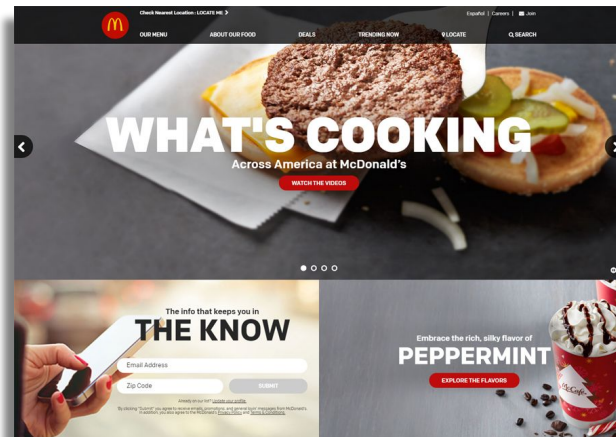
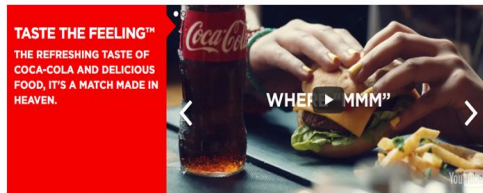
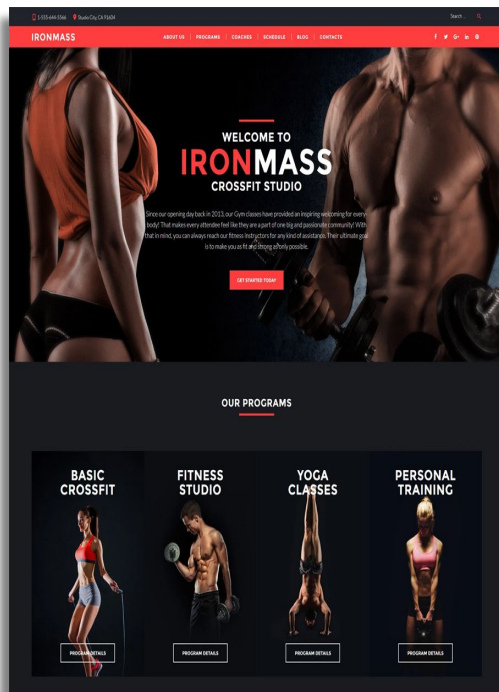




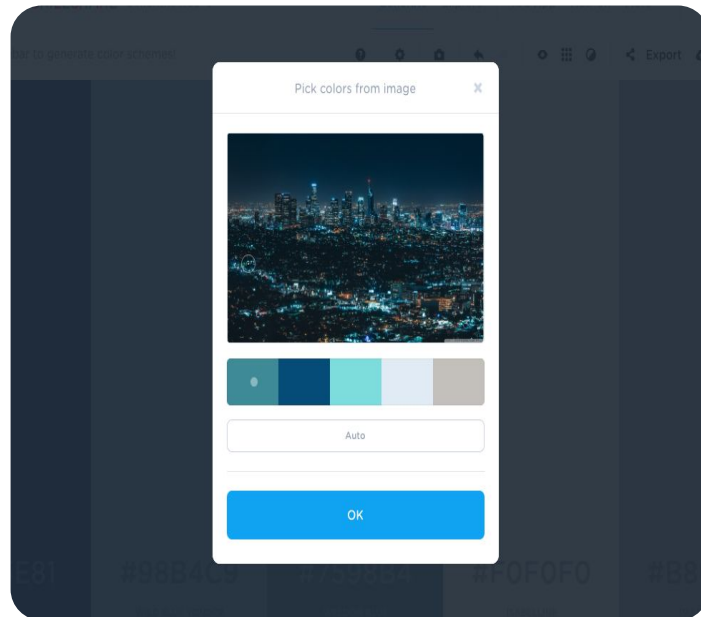
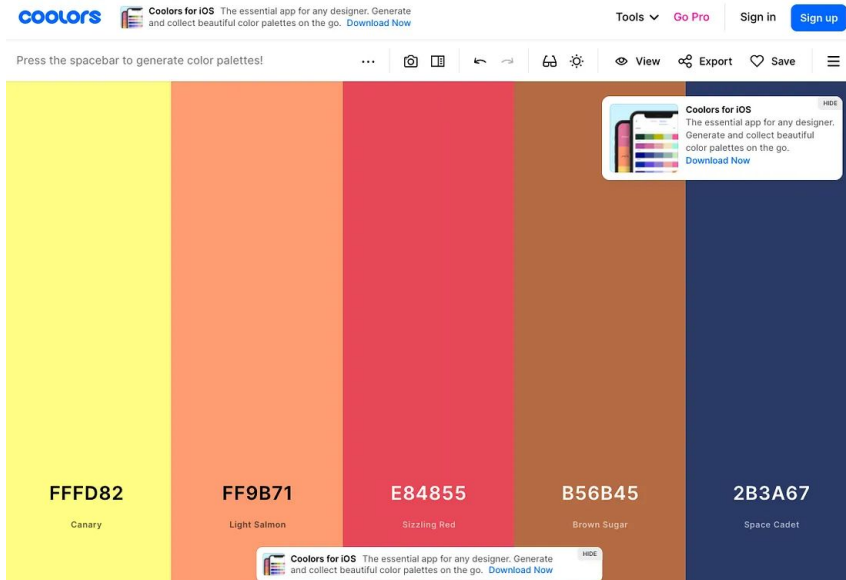
GREEN represents
Safety, nature, Wellness



USING RED – ENERGY, STIMULATE APPETISE/ danger, warning, or urgency



Tools to help you choose colours



Tools to help you choose colors



Design spotlight ▾

Business ▾

Education ▾

Plans and pricing ▾

Learn ▾



Home > Colors > Color Palette Generator

The easiest place to get colors from your photos

Want a color scheme that perfectly matches your favorite images? With Canva's color palette generator, you can create color combinations in seconds. Simply upload a photo, and we'll use the hues in the photo to create your palette.

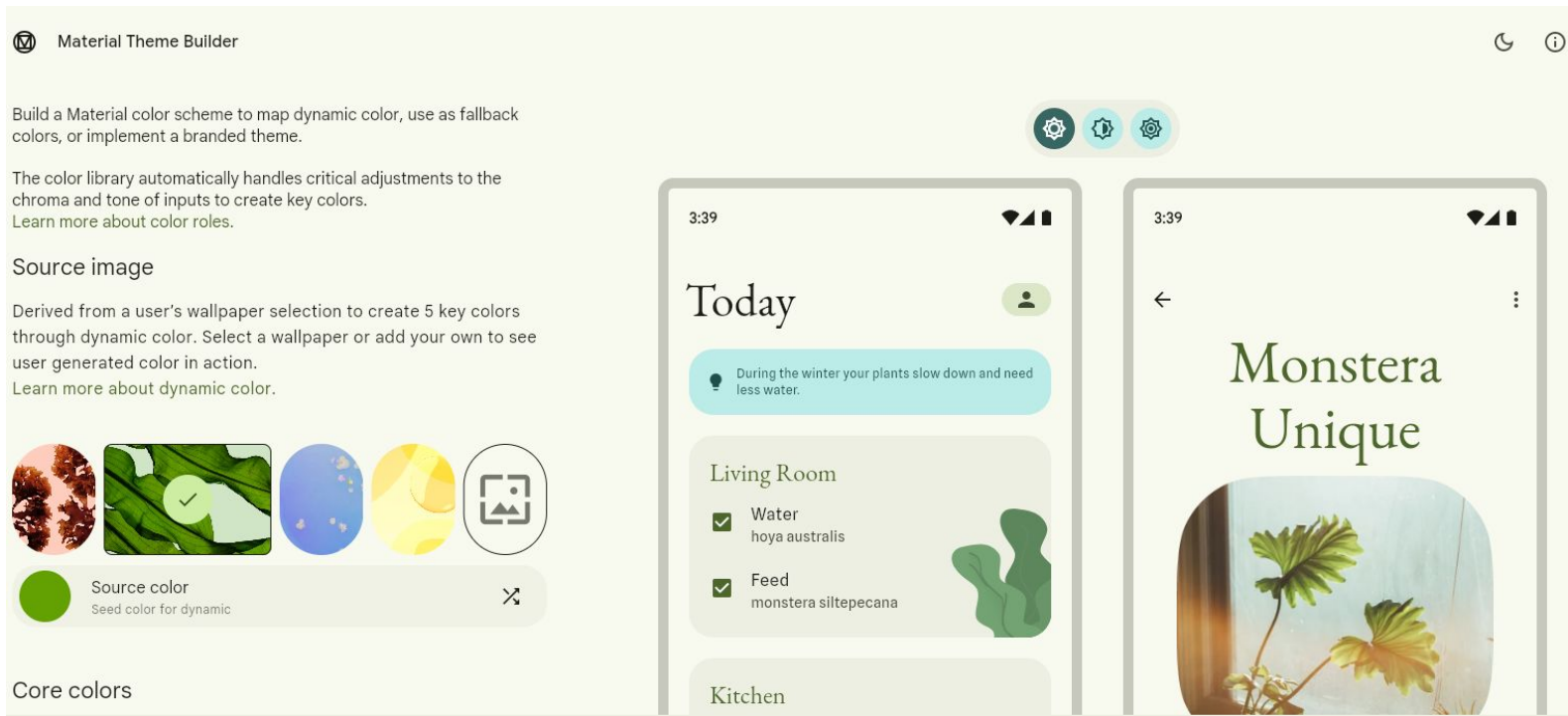
Upload an image

Try demo image



<https://www.canva.com/colors/color-palette-generator/>

Tools to help you choose colors



Dark Mode / Check Contrast

WebAIM
web accessibility in mind

services articles resources projects community

Search:

Introduction to Web Accessibility

WebAIM Training

Contrast Checker

Home > Resources > Contrast Checker

Foreground

Hex Value: #0000FF

Color Picker

Alpha: 1

Lightness

Background

Hex Value: #FFFFFF

Color Picker

Lightness

Contrast Ratio: **8.59:1**

[permalink](#)

Normal Text

WCAG AA: **Pass**

WCAG AAA: **Pass**

The five boxing wizards jump quickly.

Large Text

WCAG AA: **Pass**

WCAG AAA: **Pass**

The five boxing wizards jump quickly.

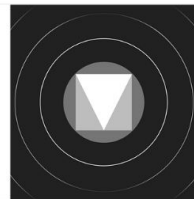
Related Resources

- Contrast and Color Accessibility
- Quick Reference: Testing Web Content for Accessibility
- WebAIM Auditing & Evaluation Services
- Web Accessibility for Designers
- Link Contrast Checker
- Contrast Checker
- Backgrounds

Material Design

Build beautiful, usable products faster. Material Design is an adaptable system-backed by open-source code-that helps...

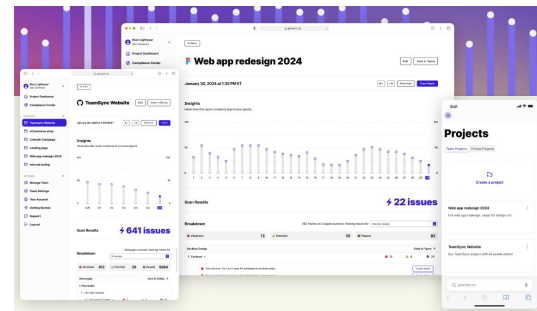
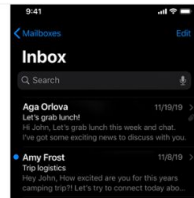
material.io



Dark Mode - Visual Design - iOS - Human Interface Guidelines - Apple Developer

In iOS 13.0 and later, people can choose to adopt a dark system-wide appearance called Dark Mode. In Dark Mode, the...

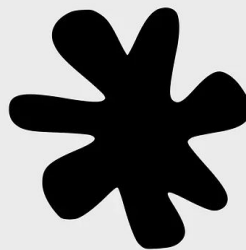
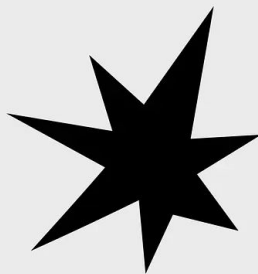
developer.apple.com



<https://webaim.org/resources/contrastchecker/>

www.getstark.co

Psychology of Shapes

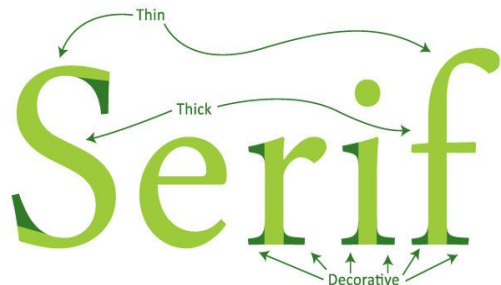


Button

Button

Button

Typography in UI



Sans Serif

No serifs
No thick/thin transition

Case Study: A Typography Issue on Mobile Screens

- Small word spacing makes text hard to read on small screens
- Usability tests showed consistently low readability scores
- Only option was to enlarge the font → but too big breaks the layout
- Typography choices directly affect usability and user satisfaction

п п



- Lowercase “l” (L) vs Uppercase “I” (i) vs Number “1”
- Lowercase “u” vs “v”
- Lowercase “rn” vs Lowercase “m”

From Flows to Prototypes miro

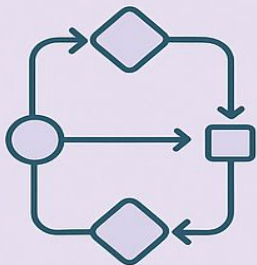
Task Flow

User Flow

Wireflow

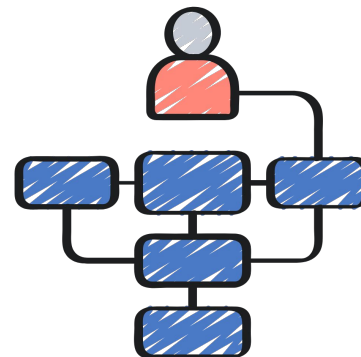
Wireframes

Prototype

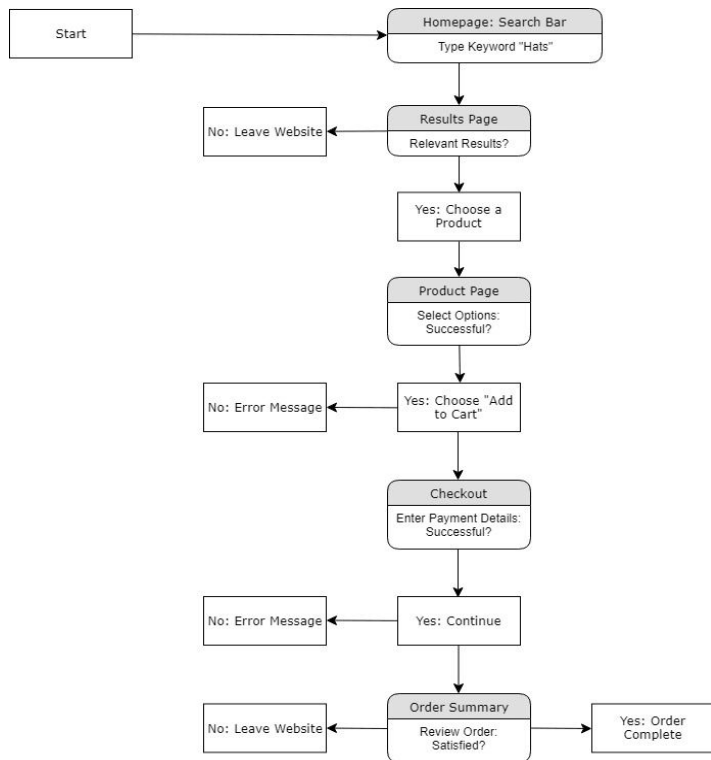


Why Start with Flow?

- Clarify the User's Journey
- Define What Screens Are Needed
- Maintain Logical Navigation
- Bridge Between Requirements and Interface
- Guide UI Layout Decisions
- Support Collaboration Across Teams
- Reveal Data Needs and Back-End Interactions



Task Flow



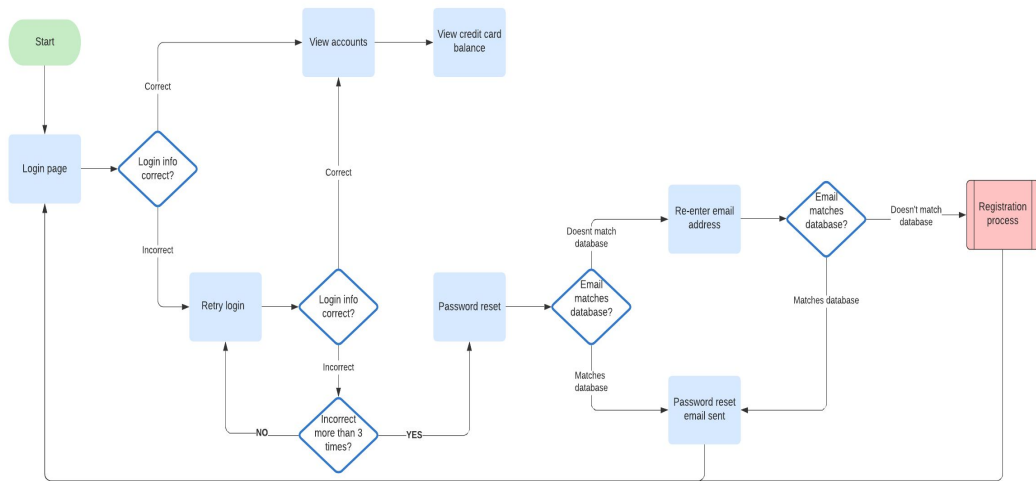
Task flows – Task flows tend to be linear, Illustrates the motivations and steps that a user takes to complete a task or accomplish a goal. A user's "Happy Path".

User Flow in UX/UI Design

User flows help determine how many screens are needed, what order they should appear in, and what components need to be present.

Why User Flows Are Important?

- Show all the possible paths a user can take to achieve (or fail) a goal.
- Help determine which screens are needed and how they connect logically.
- Serve as a blueprint for wireframes and prototypes.
- Provide a shared visual map for designers, developers, and stakeholders.



A User flow Diagram

A user flow diagram is a visual representation of the steps a user takes to complete a goal within a product (website, app, or system). In UX, there is no strict universal standard, but there are several widely accepted conventions borrowed from flowcharts and adapted for design.

◆ Shapes Used in This Course

Rectangle → Screen / Page / Step the user sees.

Home Page

Diamond → Decision point / Branch (e.g., "Login success?" → Yes/No).

Login
success?

Arrow → Flow / transition from one step to the next (user actions are written on the arrow).

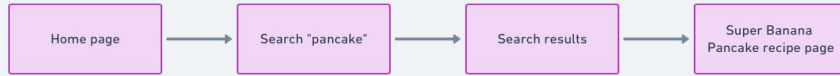
Fill out the form

Document shape → notes, errors, or messages.

Error
message

Task Flow

Task flow: finding a pancake recipe 🔍🥞

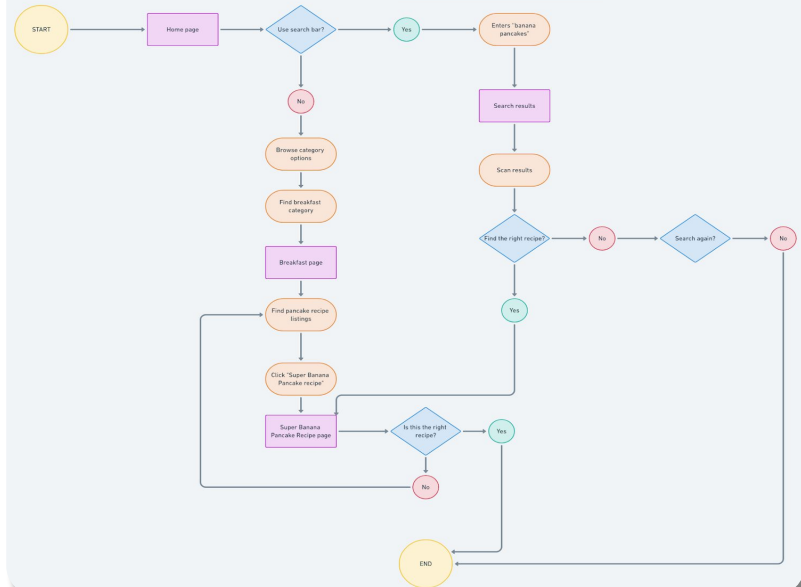


User Flow

User flow: finding a pancake recipe

Scenario: Mimi, a parent of three kids, searches for a banana pancake recipe on through the new recipe website recommended to her by a friend.

LEGEND



erika harano

Follow

Mar 3, 2019 · 4 min read ★

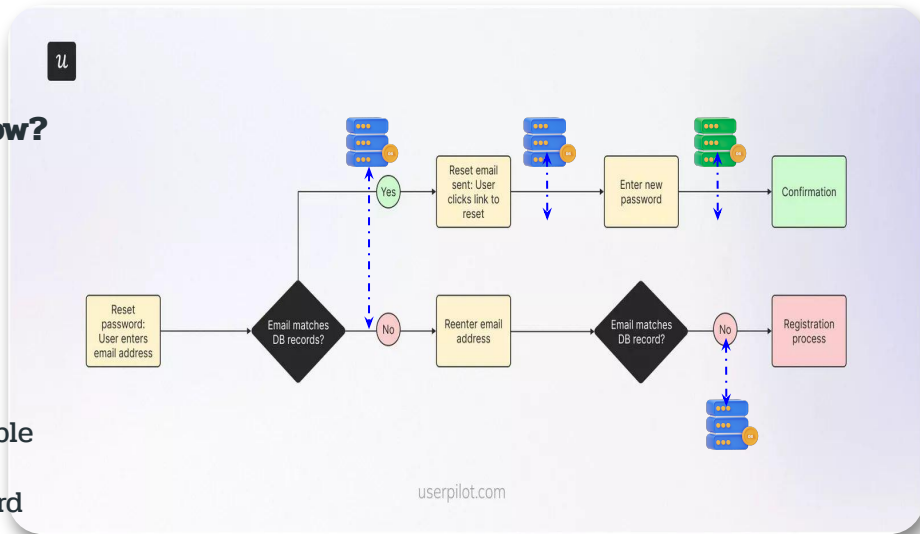
<https://medium.com/erika-harano/ux-task-flows-versus-user-flows-as-demonstrated-by-pancakes-896e78a98026#:~:text=Here's%20how%20I%20differentiate%20between,task%20flows%20and%20user%20flows.>

From User Flow to Database Thinking

What happens behind the scenes – “Reset Password” Flow?

Database Perspective

UI Step	Behind the Scenes (DB Actions)
Enter Email	Check Users table for match (Read query)
Reset Email Sent	Generate token → store in ResetTokens table
Enter New Password	Validate token → Hash and update password
Confirmation	Show success, redirect to login page



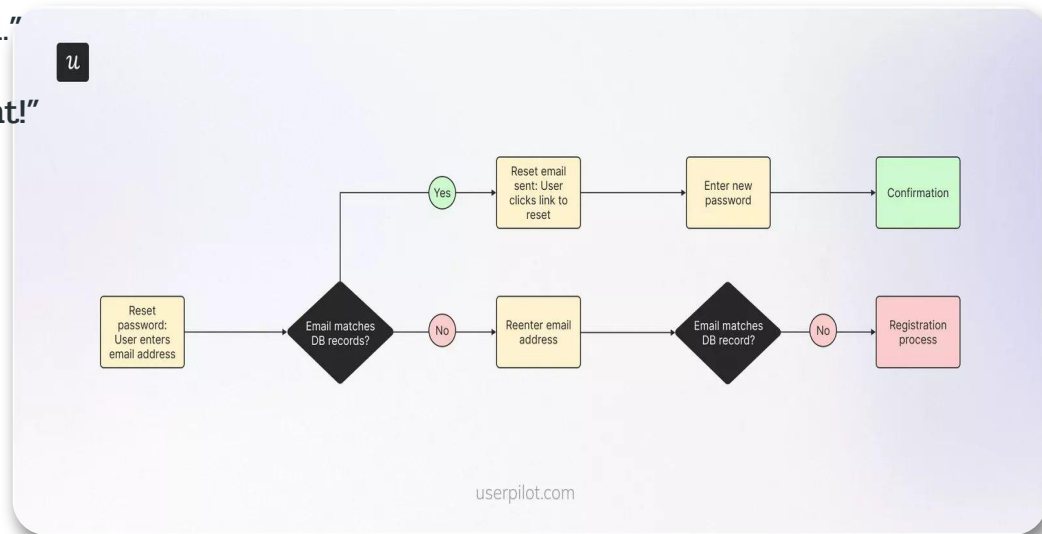
What UI Designers Must Consider?

1. State Handling in the Interface

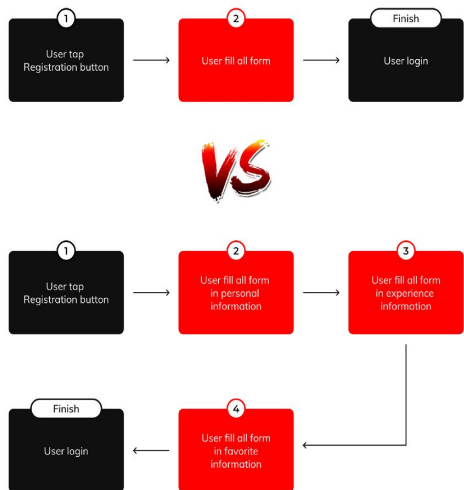
- Loading states: "Checking email..."
- Error states: "We couldn't find that email."
- Success states: "Password reset email sent!"

2. User Experience (UX) and Feedback

- Provide Clear Instructions
- Display Success and Error Messages
- Support Multiple User Scenarios



Does short task flow mean better UX?



About user flow which can be tricky

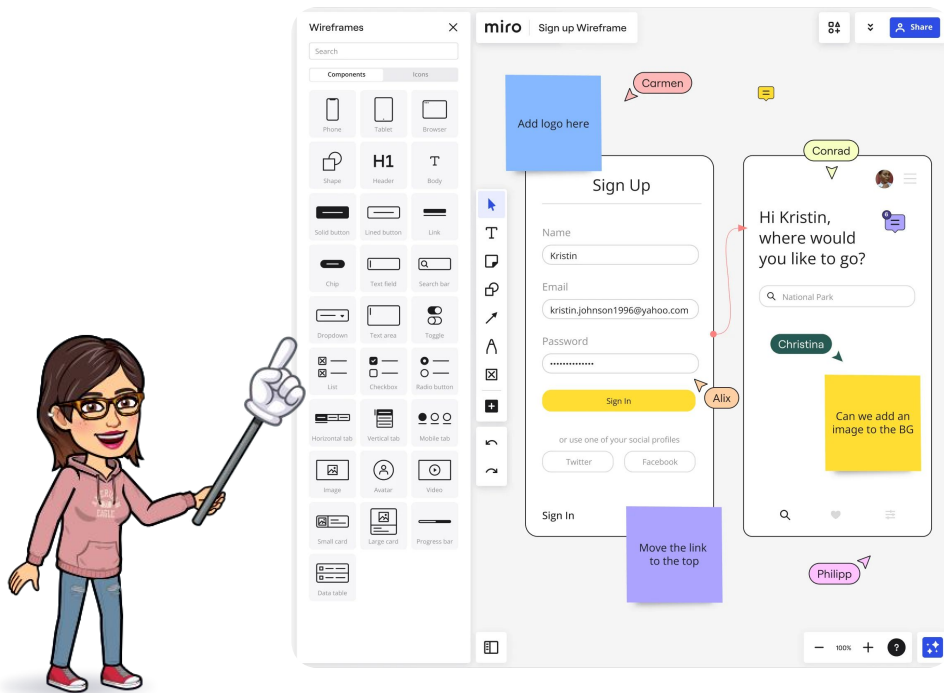


Rhasyab [Follow](#)

Dec 26, 2019 · 3 min read

Wireframe

A wireframe is a simple, low-fidelity layout that shows the structure and content placement of a screen, without styling or colors – like a blueprint for your design.



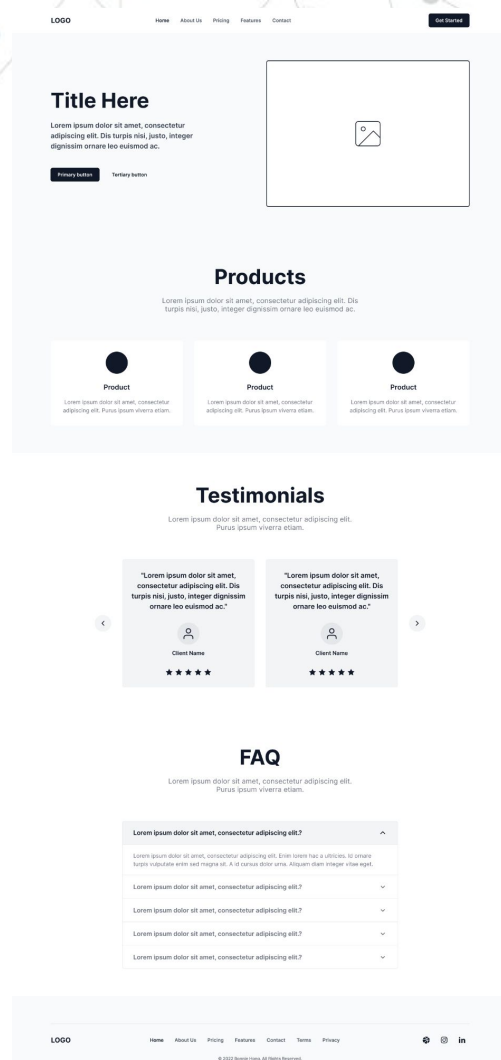
Wireframe: Low-fidelity Wireframes (Lo-fi)

- Very basic, sketch-like representations.
- Usually black & white or simple shapes.
- Focus on layout, structure, and flow (what goes where).
- Not much detail – placeholders for text (“Lorem ipsum”), grey boxes for images, lines for buttons.
- Quick to create, often hand-drawn or made with simple tools.

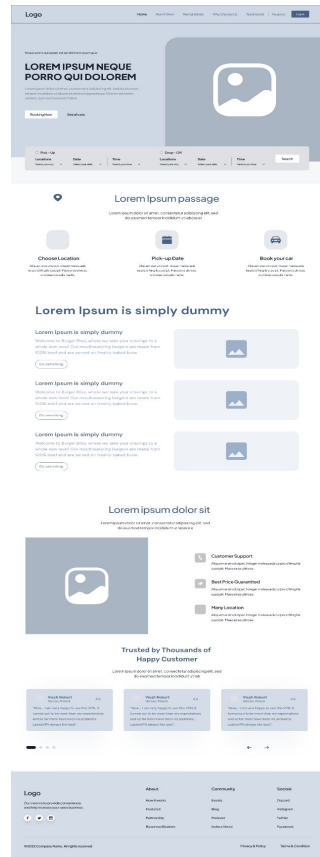
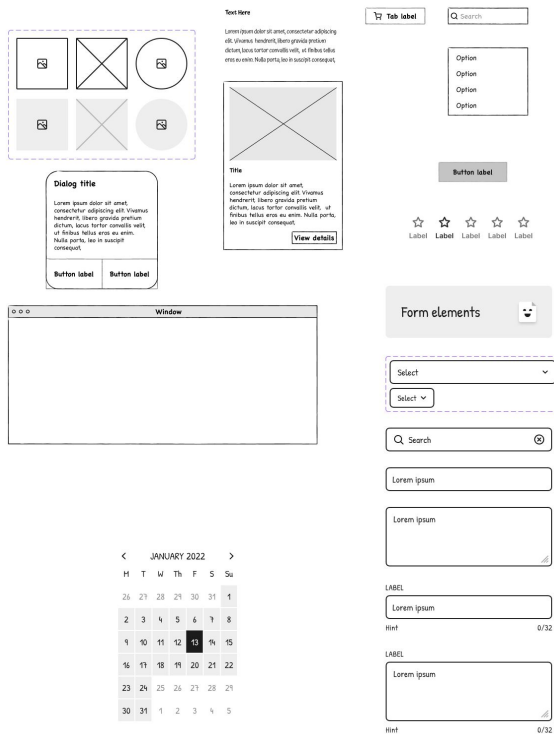
Resources:

<https://www.youtube.com/watch?v=MmkUOfE6Qqs>

<https://www.youtube.com/@MiroHQ>

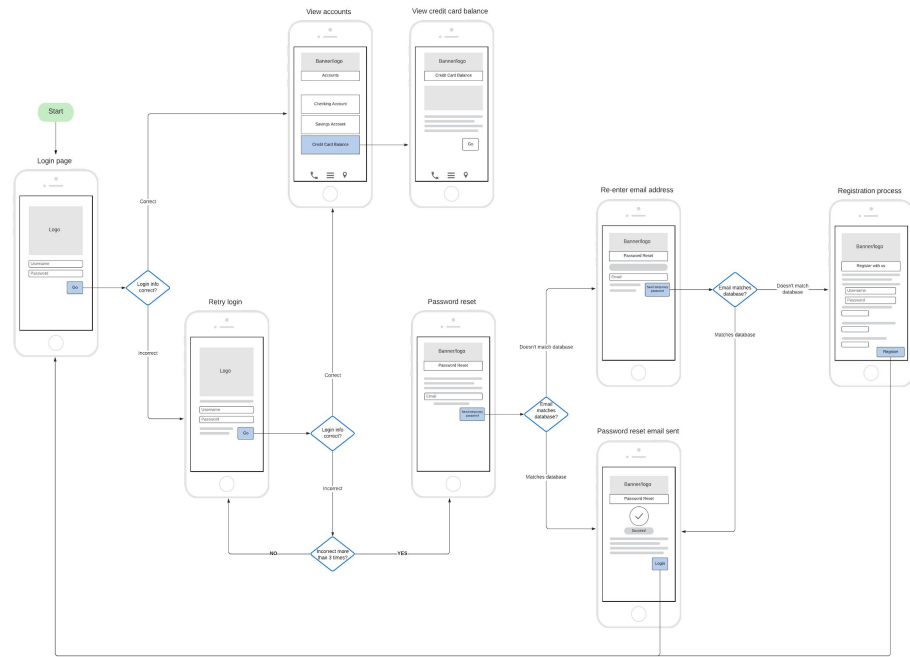


Low-fidelity Wireframes (Lo-fi) Elements



Wireflow

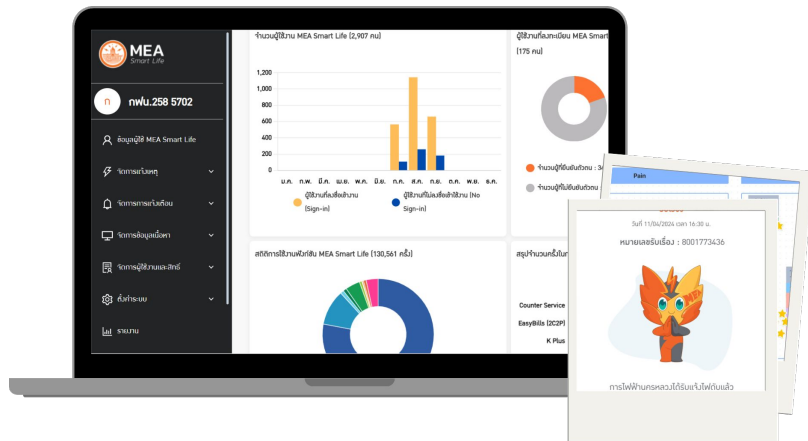
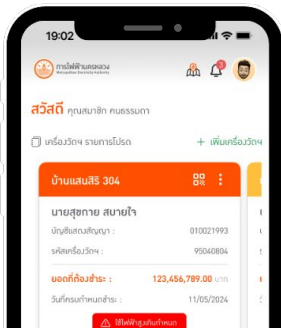
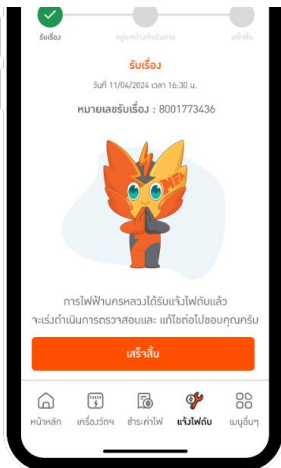
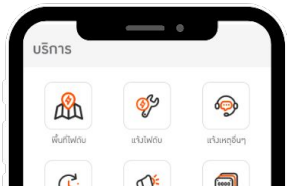
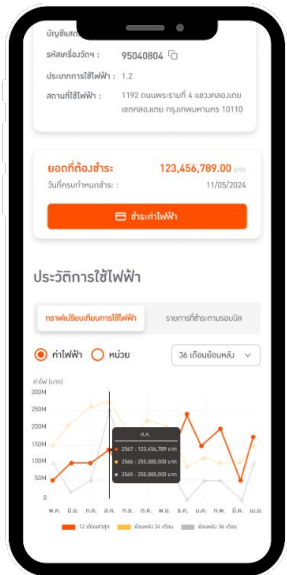
Wireflow is the representation of screen flow, by putting together a set of related wireframes following the order they appear in the flow. The use of decision (shape) in a wireflow makes it possible to present multiple navigation paths in a single flow. In short, wireflows are a combination of wireframes and user flows.



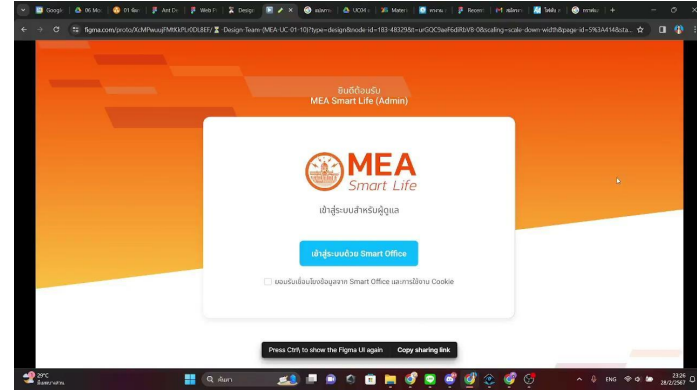
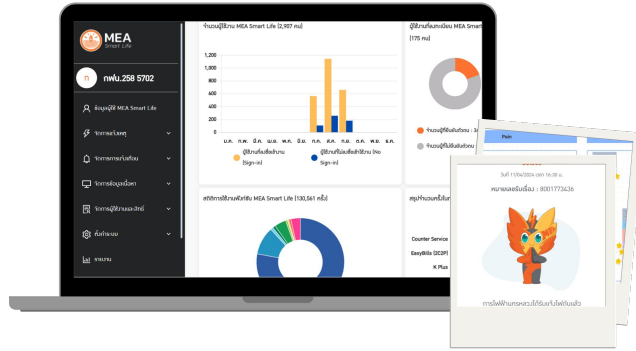


Mockup - high-fidelity wireframes (Hi-fi)

A mockup is a static, high-fidelity visual representation of a design that showcases its final look, including colors, typography, images, and branding.

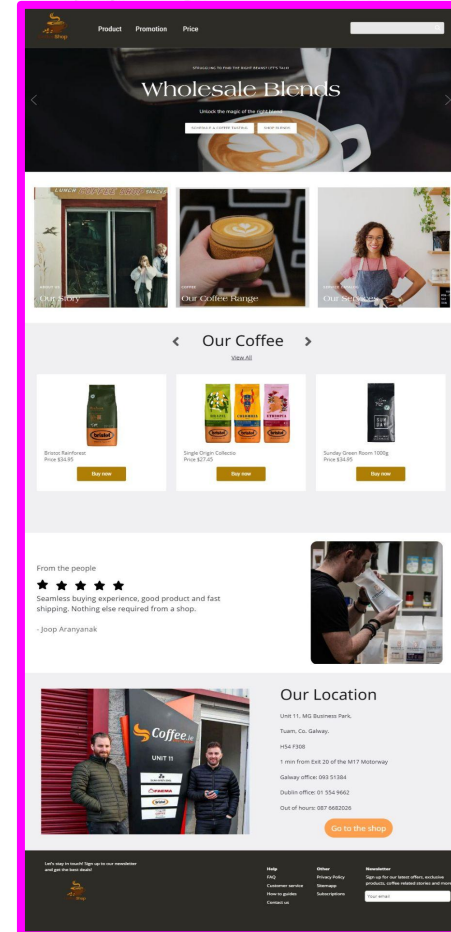
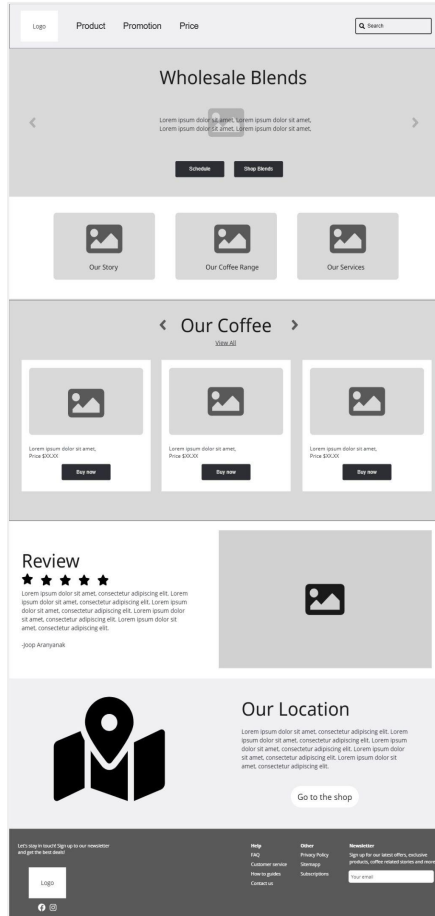


Prototype



A prototype is an interactive, high-fidelity model of a digital product that simulates real user interactions. It allows designers, developers, and stakeholders to test the user experience (UX) before full development. Unlike static wireframes or mockups, prototypes include clickable elements, transitions, and animations to mimic the final product.

<https://coffee.ie/>



Useful links about Miro

- Please watch these before the lab session.

Sign Up Miro Account - <https://www.youtube.com/watch?v=UBh0v2R8wPO>

Miro Tutorial in 3 mins - <https://www.youtube.com/watch?v=kRdtn8G4tII>

Create website wireframes - <https://www.youtube.com/watch?v=ztyZ6NSOtUg>

Miro Prototyping Tricks - <https://www.youtube.com/watch?v=ylxwibQjDk4&t=53s>

How to share Miro board - <https://www.youtube.com/watch?v=X8NxEvU0khc>

(It is recommended to watch these before the lab session, as they will help you create your work for CA1.)



Checking your attendance

Week 3 Lecture 1



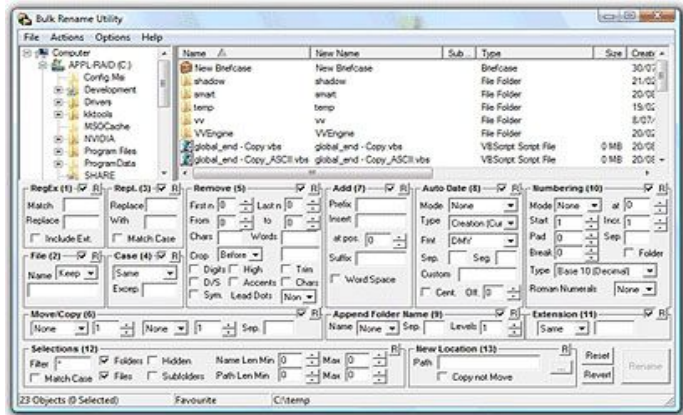
Note that you will get 1% for answering the question correctly.

End of Part 1



WEB UX DESIGN

What do you think of these UI designs – are they good or bad?



E-mail

Password

Error!

Birthday:

January 1 2017

Primary phone number*:

0 0 0 0 0 0 0 0

Secondary phone number:

0 0 0 0 0 0 0 0

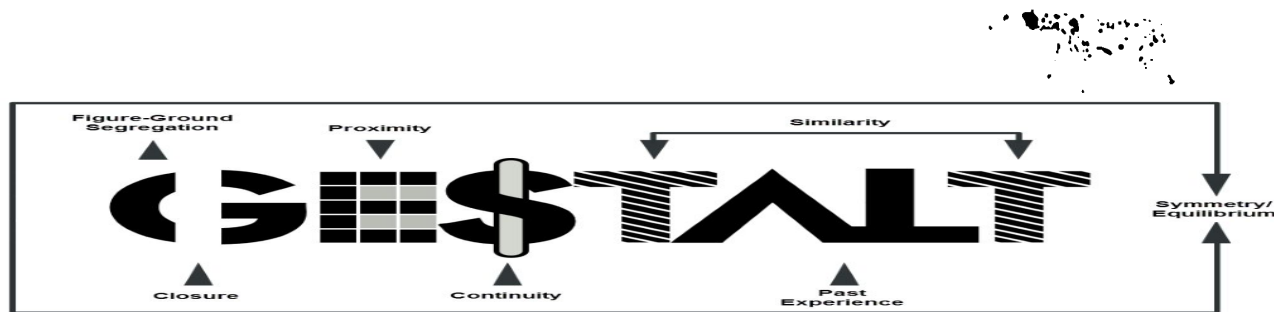
Fields v review: asterisks are required. Make sure your det
application and will email you on the progress
Make si reject y: have read and understood the terms and a
application in cases outlined in terms and agre

Graphical User Interface (GUI)

- A GUI contains graphic elements, such as windows, menus, icons, links, and other screen elements, that you select when you're working with an operating system, software application, or mobile app. A GUI enables a person to communicate with a computer through symbols, visual metaphors, and pointing devices, making interactions more intuitive and user-friendly

Gestalt theory

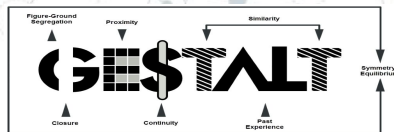
In the 1920s, a group of German psychologists developed the Gestalt Theory of visual perception, which explains how humans make sense of visual information. The term Gestalt means '**unified whole**' that refer to the way in which humans, when looking at a group of objects, will see the whole before we see the individual parts.



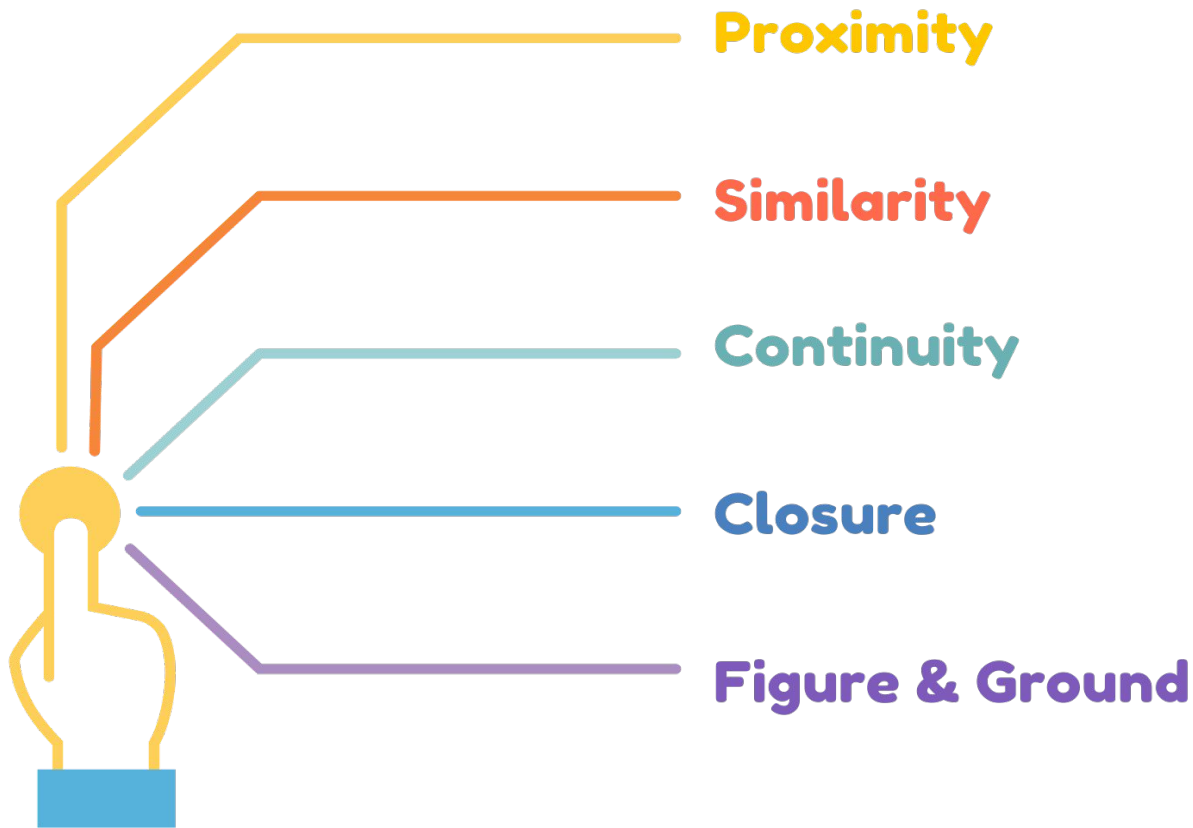
Why should designers care about the Gestalt principles?

“Great designers understand the powerful role that psychology plays in visual perception. What happens when someone’s eye meets your design creations? How does their mind react to the message your piece is sharing?”

– Laura Busche, Brand Content Strategist at Autodesk



Gestalt theory



Gestalt theory

1. **Proximity:** The law of proximity describes how the human eye perceives connections between visual elements. Elements that are close to each other are perceived to be related when compared with elements that are separate from each other.



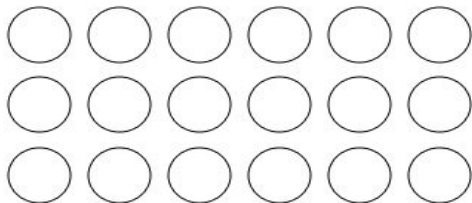
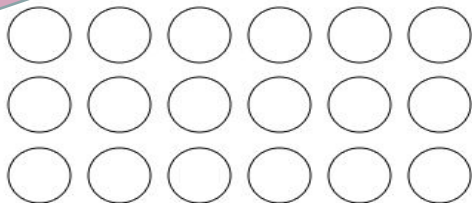
PROXIMITY

When objects placed together, the eye perceives them as a group.



Proximity

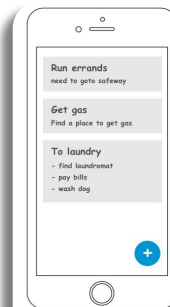
Space



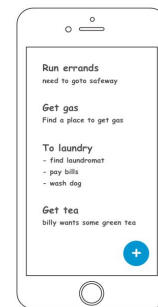
Proximity

Gestalt Principles

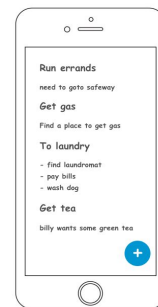
<https://uxmisfit.com>



✓ Proximity, color, and space further reinforces groupings



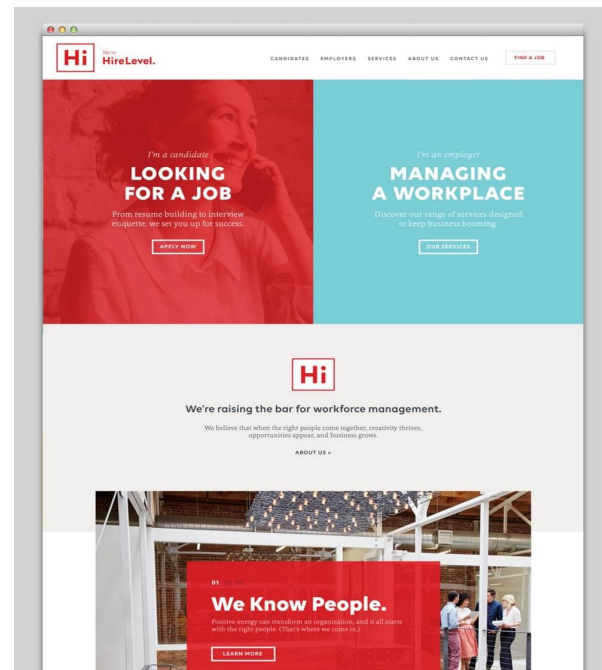
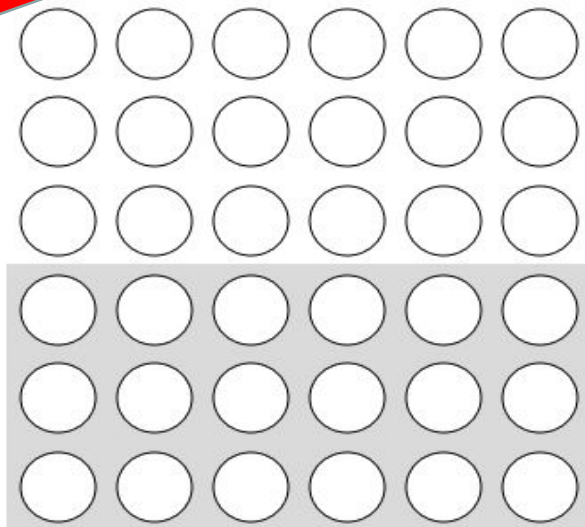
✓ Proximity and space alone can establish groups



✗ Unclear visual hierarchy when proximity isn't used for related groups

Proximity

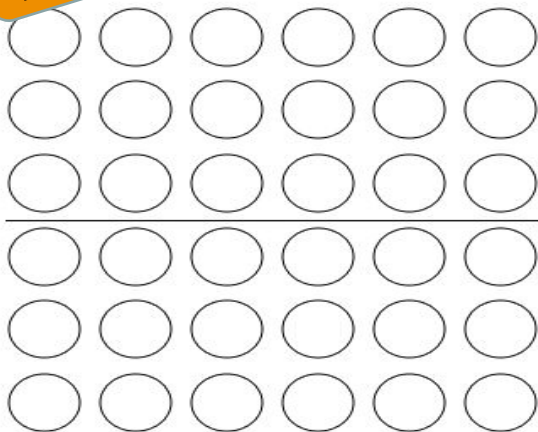
Background color



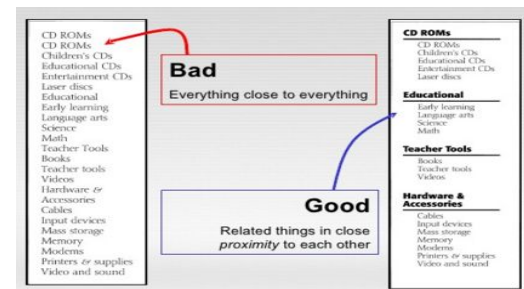
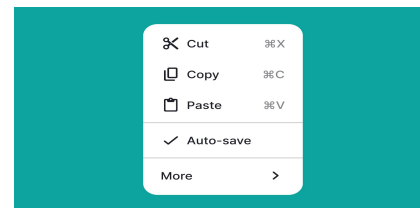
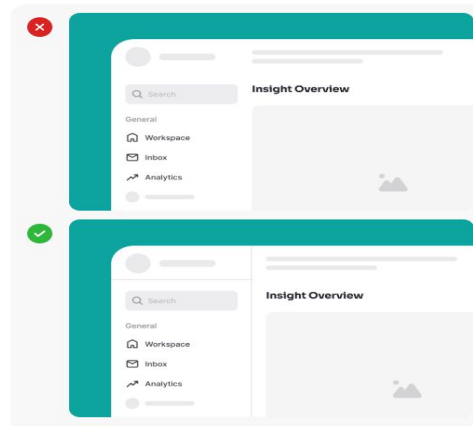
<http://www.slideshare.net/jaeger33/carp-design-principles>,
<https://app.uxcel.com/courses/ui-components-best-practices/basics-676>

Proximity

Line



Vertical line divider



Gestalt theory

2. **Similarity:** The principle of similarity states that things which share visual characteristics such as shape, size, color, texture, or orientation are perceived as belonging together.



Similarity

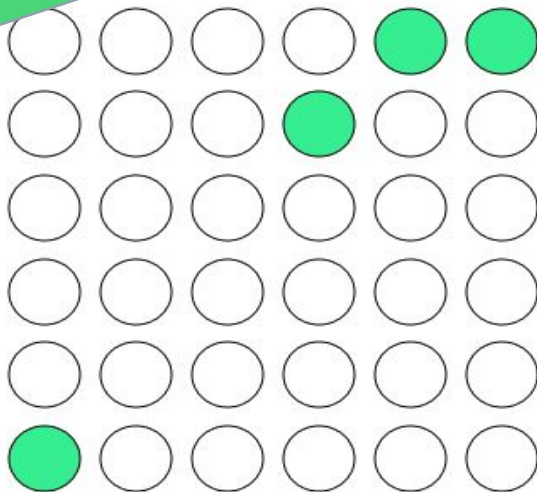


Gestalt Principles

<https://uxmisfit.com>

Similarity

Color



UI Blogazine is the first UI blog in Thailand

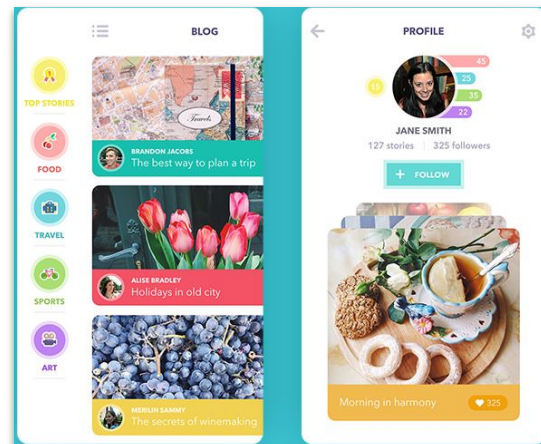
Author: [Furinto](#)

This is Photoshop's version of Lorem Ipsum. Proin gravida nibh vel velit auctor aliquet. Aenean sollicitudin, lorem quis bibendum auctor, nisi elit consequat ipsum, nec sagittis sem nibh id elit. [Duis sed](#) odio sit amet nibh vulputate cursus a sit amet mauris. Morbi accumsan ipsum velit. Nam nec tellus a odio tincidunt auctor a ornare odio. Sed non mauris vitae erat consequat auctor eu in elit. Class aptent taciti sociosqu ad

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[Great Designers](#)
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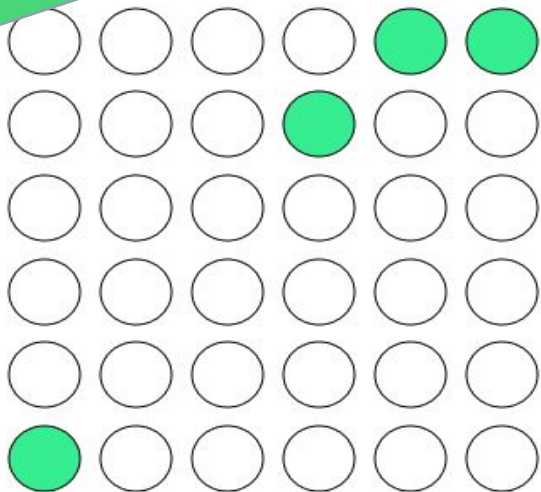


<http://www.uiblogazine.com/ui-theory/gestalt-for-uid>

<https://design4users.com/gestalt-theory-for-ux-designers-principle-of-similarity/>

Similarity

Color



Similarity

Product Name
Description and features

Edit

Remove

Product Name
Description and features

Edit

Remove

Product Name
Description and features

Edit

Remove

Product Name
Description and features

Edit

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Product Name
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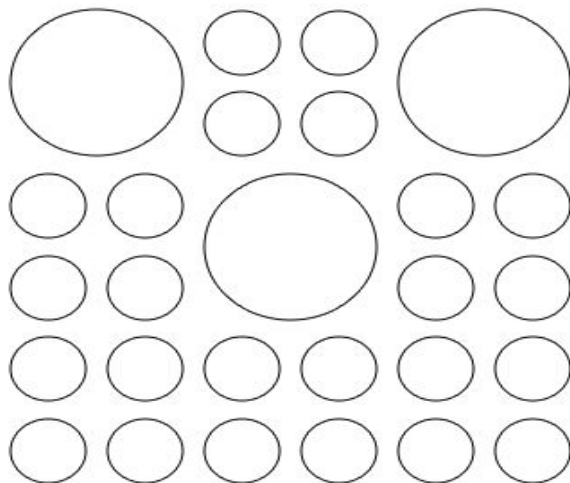


Gestalt Principles

<https://uxmisfit.com>

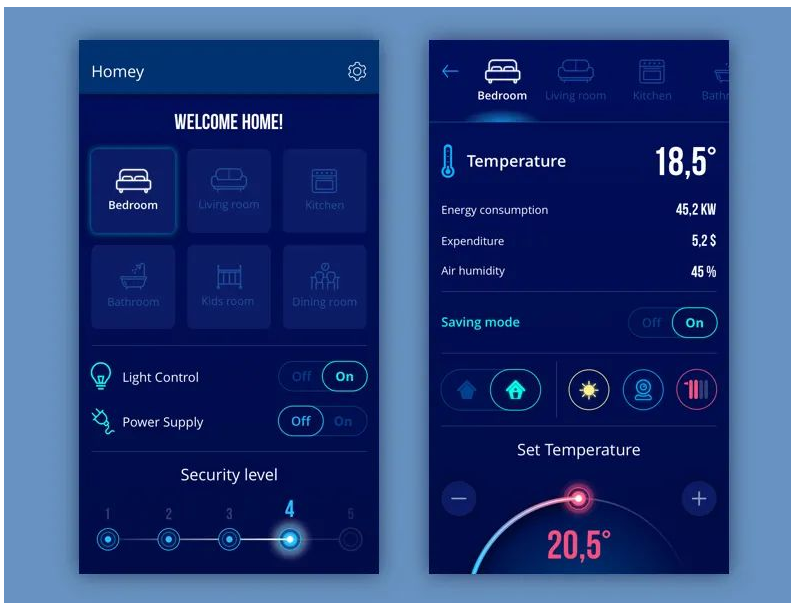
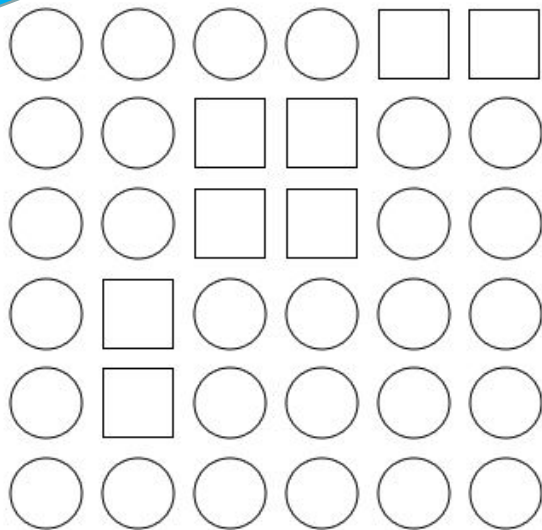
Similarity

Size



Similarity

Shape



Proximity + Similarity in UI design

Appliances
Audio, TV & Home Theatre
Blu-ray
Camera & Photo
Computers & Office
DVD
Electronics
Garden & Outdoors
Home & Garden
Houseware & Furnishings
iPod, MP3 & Accessories
Kitchen & Dining
Lighting
MP3 Downloads
Music
Music, DVD & Games
Office Products & Supplies
PC & Video Games
PCs & Laptops
Peripherals & Accessories
Sat Nav, Phones & PDA
Software

Music, DVD & Games

Music
MP3 Downloads
DVD
Blu-ray
PC & Video Games

Electronics

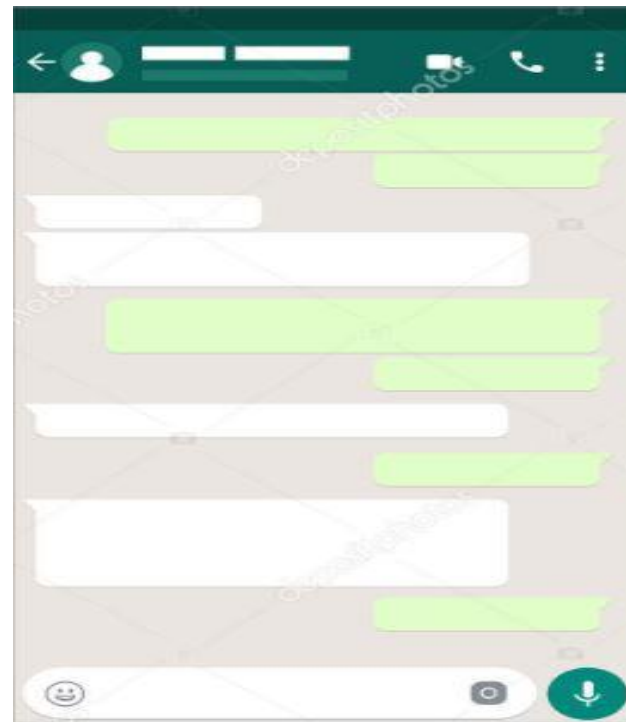
Camera & Photo
Audio, TV & Home Theatre
iPod, MP3 & Accessories
Sat Nav, Phones & PDA

Computers & Office

PCs & Laptops
Peripherals & Accessories
Software
Office Products & Supplies

Home & Garden

Kitchen & Dining
Appliances
Houseware & Furnishings
Lighting
Garden & Outdoors

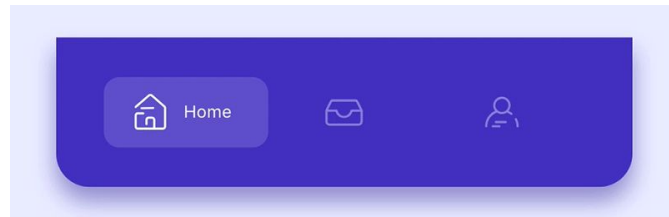
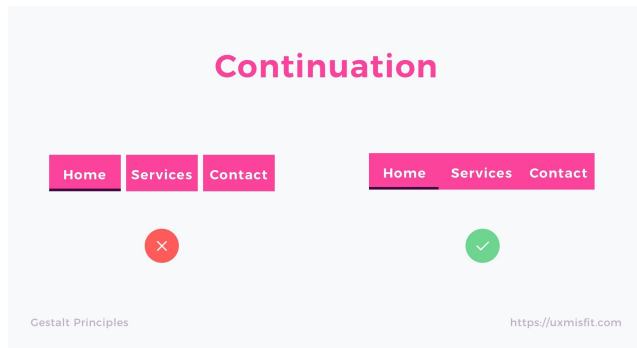


Gestalt theory

3. **Continuity**: Law explains how our brain experiences visual line of elements that are grouped together. There is a tendency to perceive a line continuing its established direction.



<https://e-strategy.com/2012/08/13/sunsilk-print-ad-eyetracking-heatmap/>



<https://medium.muz.li/gestalt-principles-in-ui-design-6b75a41e9965>

Gestalt theory

4. **Closure:** Gestalt psychologists believe that the brain tends to perceive forms and figures in their complete appearance despite the absence of one or more of their parts, either hidden or totally absent.



Gestalt theory

การออกแบบกราฟิกตามหลักการของ Gestalt theory หรือทฤษฎีการรับรู้ (Perception) นั้นมีพื้นฐานมาจากหลักการที่ว่า มนุษย์จะรับรู้ภาพตามลักษณะที่ปรากฏ ไม่ใช่ตามลักษณะที่แท้จริง ตัวอย่างเช่น หากเราเห็นภาพของต้นไม้ที่มีกิ่งก้านและใบ เราจะไม่รับรู้ว่าเป็นภาพของต้นไม้ที่มีกิ่งก้านและใบ แต่เราจะรับรู้ว่าเป็นภาพของต้นไม้ที่มีกิ่งก้านและใบที่มีลักษณะเฉพาะตัว

หลักการของ Gestalt theory มีดังนี้:

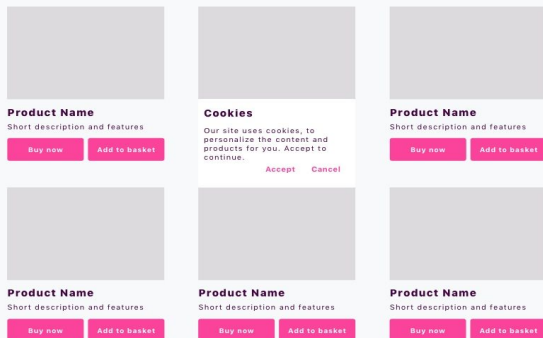
- 1. ความต่อเนื่อง (Continuity): มนุษย์จะรับรู้ภาพตามลักษณะที่ปรากฏอย่างต่อเนื่อง
- 2. ความคล้ายคลึง (Similarity): มนุษย์จะรับรู้ภาพตามลักษณะที่คล้ายคลึงกัน
- 3. ความใกล้ชิด (Proximity): มนุษย์จะรับรู้ภาพตามลักษณะที่ใกล้ชิดกัน
- 4. ความเรียบง่าย (Simplicity): มนุษย์จะรับรู้ภาพตามลักษณะที่เรียบง่าย
- 5. ความสมดุล (Balance): มนุษย์จะรับรู้ภาพตามลักษณะที่สมดุล

การออกแบบกราฟิกตามหลักการของ Gestalt theory นั้นมีประโยชน์อย่างมากในการออกแบบกราฟิกที่สามารถดึงดูดสายตาของผู้ชมได้อย่างมีประสิทธิภาพ

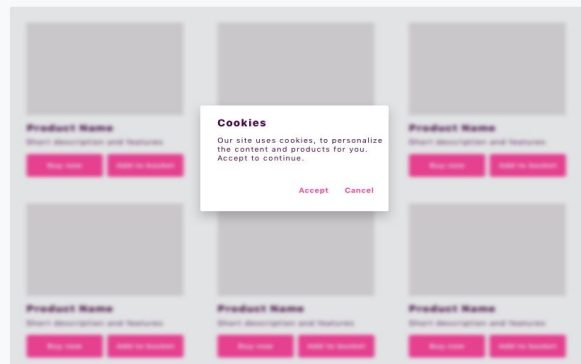
5. **Figure and Ground:** we tend to segment our visual world into figure and ground. Figure is the object or person that is the focus of the visual field, while the ground is the background.



Figure and Ground

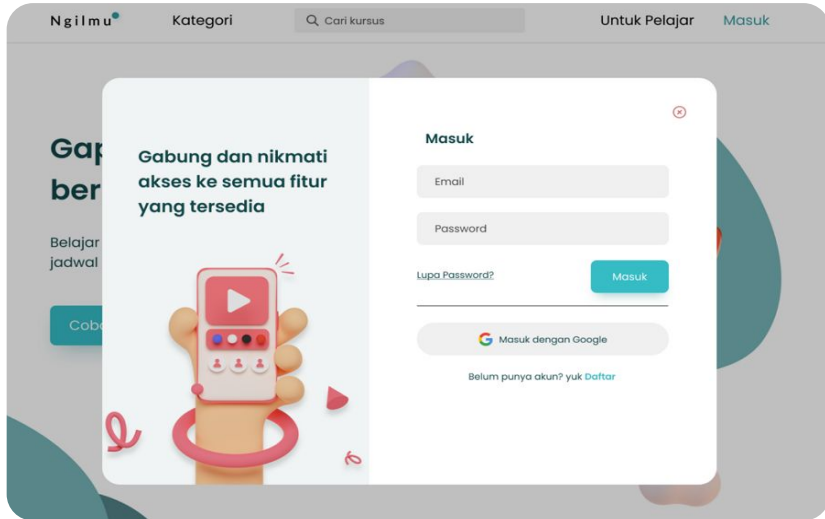
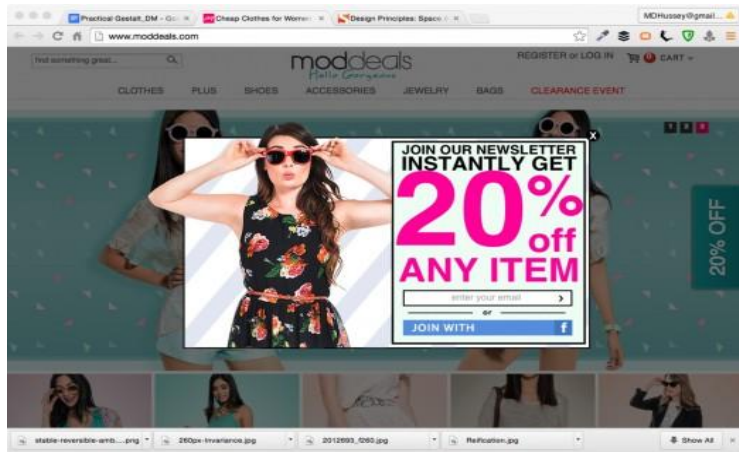


Gestalt Principles



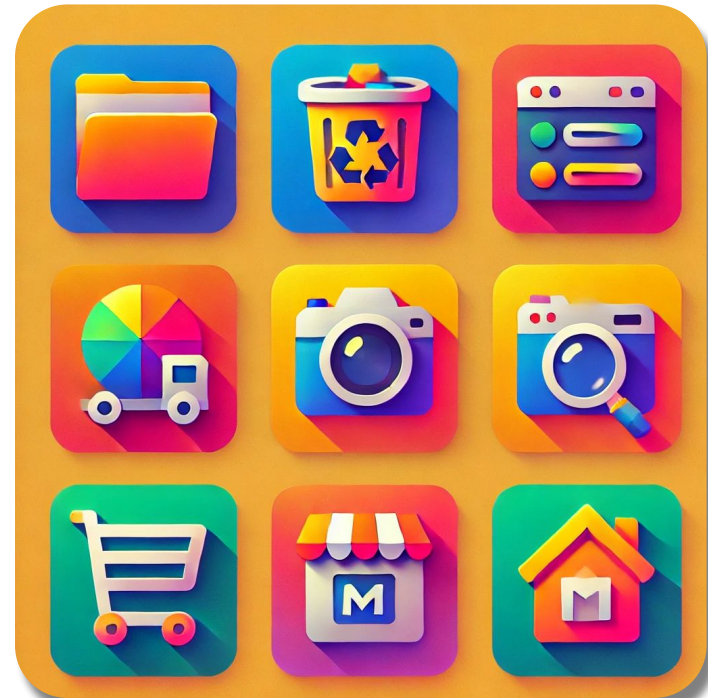
<https://uxmisfit.com>

Figure and Ground

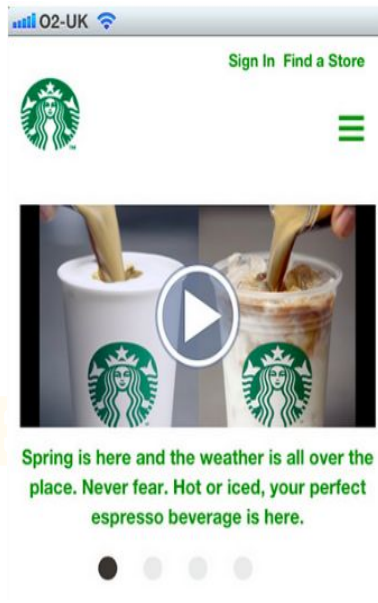
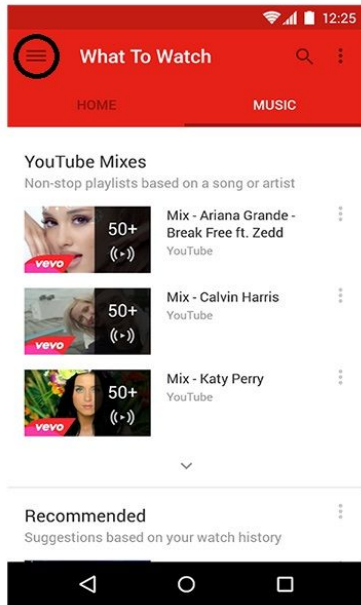


Metaphor in UI Design

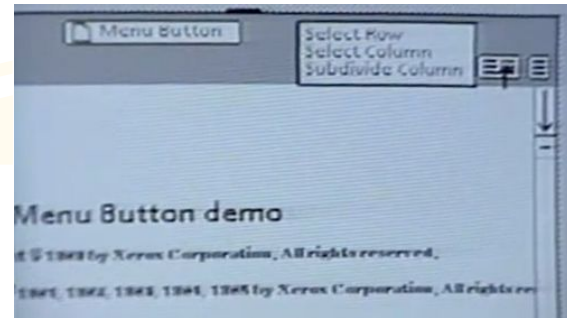
A metaphor in UI (User Interface) design is a design principle that uses familiar real-world objects, concepts, or actions to help users understand digital interactions more easily. By leveraging users' prior knowledge, metaphors make interfaces more intuitive and reduce the learning curve.



What is the name of this Icon?

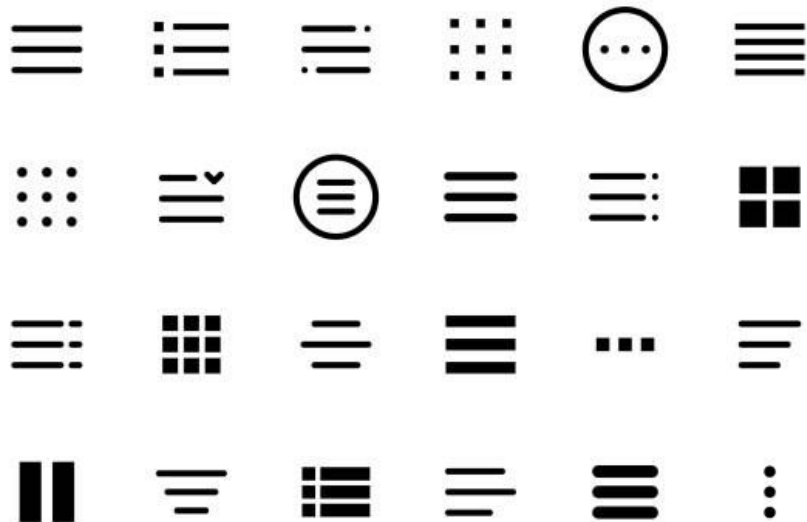


Norm Cox



Xerox Star (1981)

Menu List

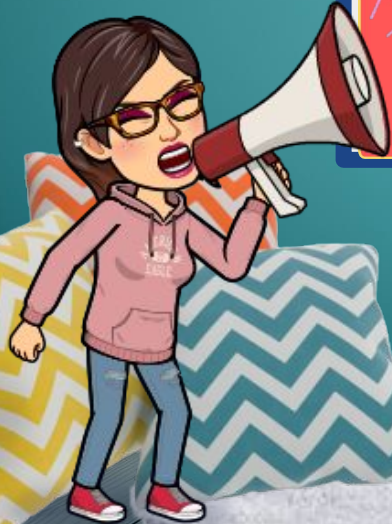
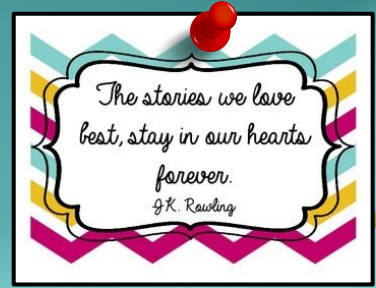
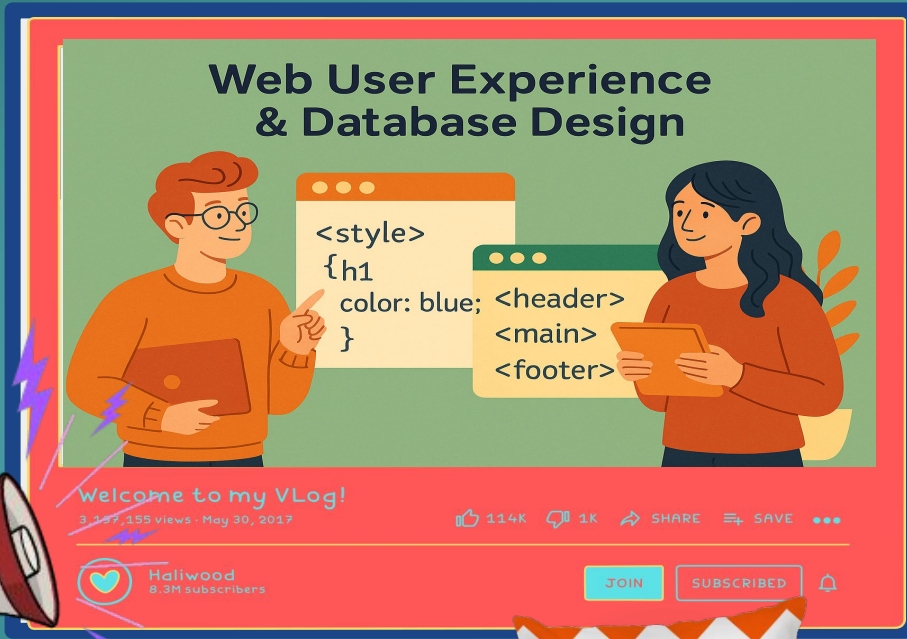




Checking your attendance Week 3 Lecture 2



Note that you will get 1% for answering the question correctly.



Contact me

You can contact me via :



Inthraporn.aranyanak@tudublin.ie

